20/2/26 Experiment: 3

Study of Different types of Network cables & crimping of cable with RJ45 connector.

Aim:

To study different types of Network cables and crimping of cable with RJ45 connector.

Cable Types:

1. UTP – Unshield Twisted Pair (UTP) cable
   - Category 3 – 10 Mbps
   - Category 5 – upto 100 Mbps
   - Category 5e – upto 1 Gbps (Improved Performance)

Adva: * cheaper, easy to install, smaller diameter.
* more prone to EMI and noise.

2. STP – shielded Twisted Pair (STP) cable
   - Category 6/6a – upto 10 Gbps
   - Category 7 – upto 10 Gbps

3. Co axial cable:
   RG-6, RG-59, RG-11 – 10-100 Mbps

4. Fiber optic cable:
   Single mode, Multi-mode – upto 100 Gbps

Note:

1. UTP is the most commonly used cable in LAN networks.
2. STP has additional shielding to protect against electromagnetic interfabce.
3. Coaxial cable consists of a central conductor surrounded by insulation and shielding

4. Fibre optic cable transmits data using light signals & provides very high speed and bandwidth

Crimping of Network cable with RJ45 Connector

RJ45 color code Arrangement

Pin No	Wire Colors
1	white-green
2	green
3	white-orange
4	Blue
5	white-Blue
6	orange
7	white-Brown
8	Brown

Step 1: To start construction of the device, begin by threading.

Step 2: Next, strip appox 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3: After, will need to untangle the wires; there should be four "twisted pairs". Reflecting back to the shot, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4: Once the cable is collect, bunch them together in a line, and if there are any that stick out further than others, chip them back up the ends.

Step 5: Next push the cable right in. The notch at the end of the plug, need to be just over the cable shielding, and if isn't it, that means that you stripped off the much shielding, simply strip the cable back a little more.

Step 6: After the wire are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7: Lastly, repeat for the other end using diag. B to make a crossover cable using diagram A.

Result:

Thus the types of Network cables are studied and crimped the cable with RJ45 connector.